

Fig. 5: An actual-size, single-side PCB layout for the display section

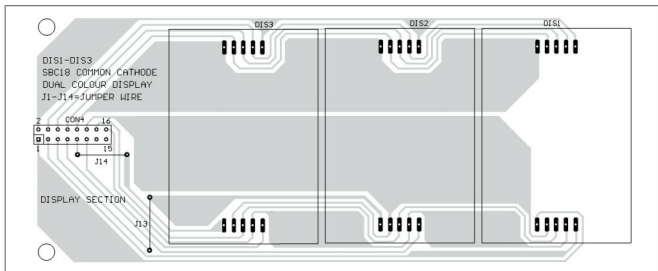


Fig. 6: Component layout for the PCB in Fig. 5

the display section is shown in Fig. 5 and its component layout in Fig. 6. Connect switches S1 and S2 to the red and green signal input lines for manual testing of the circuit.

Grid and segment control signals are fed to the display section through a 16-pin, dual-line, 2.54mm berg strip type female connector provided on the PCB of the display section. Note that the RTC section is not included in the PCB.

Make sure that all the components are mounted and wiring done as per the schematic diagrams. At power-on, the timer unit reads the red and

green traffic signals. It learns the time after two cycles of red-green signaling and the next cycle onwards, the count-down time is displayed in the same colour as the traffic signal light. That is, when the traffic light is green the count-down time too is displayed in green colour, and when the traffic light is red the timer display is in red.

Before connecting the traffic light signals into the circuit, manually test the circuit for proper working as follows: Set the red-colour timer display by keeping switch S1 pressed until the desired time, say, 10 counts, is reached. Release S1 and then press

it again until the display shows 10 counts. Similarly, for the green-colour timer display, press switch S2 until the desired time, say, 7 counts. Release it and then press again until the display shows 7 counts. Now if you press S1 again and hold it in this position, you will see the count-down in red colour from 10 to 0. Next, if you press S2 and hold it in this position, you will see the count-down in green colour from 7 to 0. This means that the system is working well. Now you can connect the red and green signal lines to respective points of CON2 as shown in Fig. 1.